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Roll no: SU92-BSSEM-F25-121

Section: 2B

PF Assignment

Program 1

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int R, C;
```

```
    cin >> R >> C;
```

```
    int arr[R][C];
```

```
    int sum = 0;
```

```
    // Input array elements
```

```
    for (int i = 0; i < R; i++)
```

```
    {
```

```
        for (int j = 0; j < C; j++)
```

```
        {
```

```

        cin >> arr[i][j];

        sum = sum + arr[i][j];
    }
}

```

```

// Output total sum

```

```

cout << sum;

```

```

return 0;

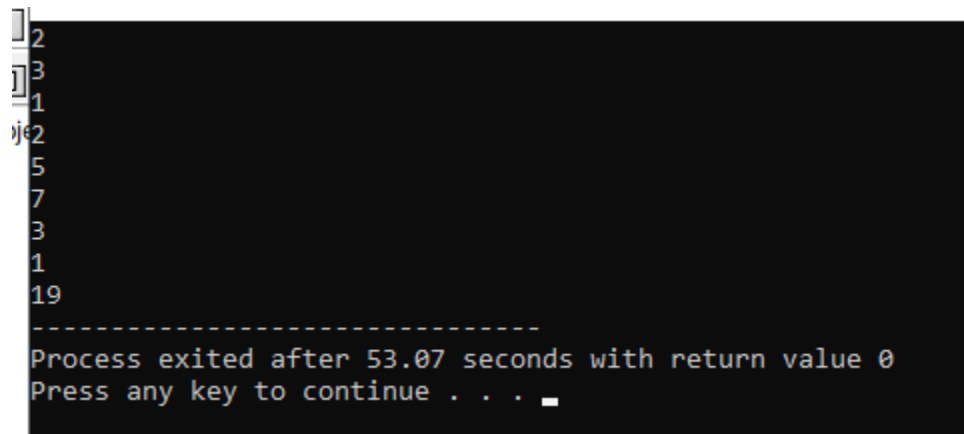
```

```

}

```

Output



```

2
3
1
4
5
7
3
1
19
-----
Process exited after 53.07 seconds with return value 0
Press any key to continue . . .

```

Program 2

```

#include <iostream>

```

```

using namespace std;

```

```

int main()

```

```
{  
    int R, C;  
  
    cin >> R >> C;  
  
    int arr[R][C];  
  
    // Input array elements  
    for (int i = 0; i < R; i++)  
    {  
        for (int j = 0; j < C; j++)  
        {  
            cin >> arr[i][j];  
        }  
    }  
  
    // Start with the first element  
    int result = arr[0][0];  
  
    // Subtract all remaining elements  
    for (int i = 0; i < R; i++)  
    {  
        for (int j = 0; j < C; j++)
```

```

    {
        if (i == 0 && j == 0)
            continue;

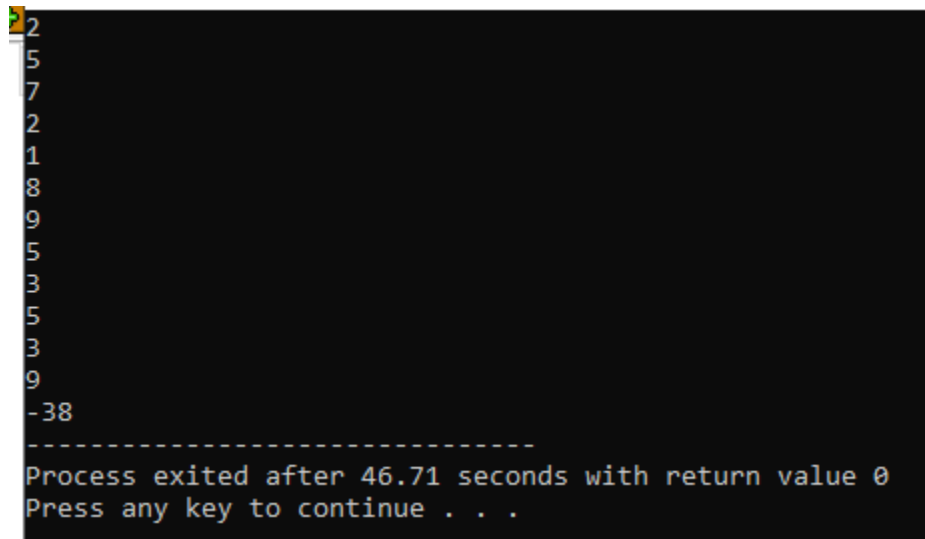
        result = result - arr[i][j];
    }
}

// Output the final result
cout << result;

return 0;
}

```

Output



```

2
5
7
2
1
8
9
5
3
5
3
9
-38
-----
Process exited after 46.71 seconds with return value 0
Press any key to continue . . .

```

Program 3

```
#include <iostream>

using namespace std;

// Function for square
double calculateArea(double side)
{
    return side * side;
}

// Function for rectangle
double calculateArea(double length, double width)
{
    return length * width;
}

int main()
{
    double side, length, width;

    cin >> side;

    cin >> length >> width;
```

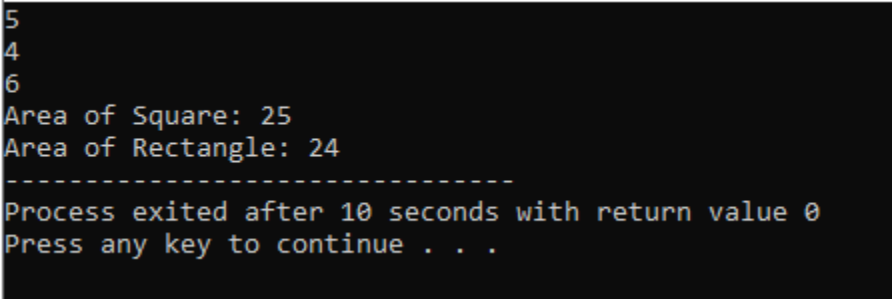
```
cout << "Area of Square: " << calculateArea(side) << endl;

cout << "Area of Rectangle: " << calculateArea(length, width);


return 0;

}
```

Output



```
5
4
6
Area of Square: 25
Area of Rectangle: 24
-----
Process exited after 10 seconds with return value 0
Press any key to continue . . .
```

Program 4

```
#include <iostream>

using namespace std;


// Function to calculate bowling average

double getBowlingAverage(int runsConceded, int wicketsTaken)

{
    if (wicketsTaken == 0)
    {
        return 0.0;
    }
}
```

```

    }

    return (double)runsConceded / wicketsTaken;
}

int main()
{
    int runsConceded, wicketsTaken;

    cin >> runsConceded >> wicketsTaken;

    cout << "Bowling Average: " << getBowlingAverage(runsConceded,
wicketsTaken);

    return 0;
}

```

Output

```

120
4
Bowling Average: 30
-----
Process exited after 10.05 seconds with return value 0
Press any key to continue . . .

```

Program 5

```
#include <iostream>

using namespace std;

// Function to print each character on a new line
void printVertical(char word[], int size)
{
    for (int i = 0; i < size; i++)
    {
        cout << word[i] << endl;
    }
}

int main()
{
    char word[100];
    int size;

    cin >> word;
    cin >> size;

    printVertical(word, size);

    return 0;
```



```
}
```

Output

```
HELLO
S
H
E
L
L
O

-----
Process exited after 22.26 seconds with return value 0
Press any key to continue . . .
```

Program 6

```
#include <iostream>

using namespace std;

// Function to reverse the word
void reverseWord(char word[], int size)
{
    for (int i = 0; i < size / 2; i++)
    {
        char temp = word[i];
        word[i] = word[size - 1 - i];
        word[size - 1 - i] = temp;
    }
}
```

```
}
```

```
int main()
```

```
{
```

```
    char word[100];
```

```
    cin >> word;
```

```
    int size = 0;
```

```
    while (word[size] != '\0')
```

```
    {
```

```
        size++;
```

```
    }
```

```
    reverseWord(word, size);
```

```
    cout << "Reversed Word: " << word;
```

```
    return 0;
```

```
}
```

Output

HELLO

Reversed Word: OLLEH

Process exited after 10.96 seconds with return value 0

Press any key to continue . . . █